

Russell Rosa

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EDUCATION

Purdue University, College of Science

West Lafayette, IN

Bachelor of Science, Data Science and Computer Science GPA: 3.3

May 2026

PROFESSIONAL EXPERIENCE

Peterson Technology Partners

Park Ridge, IL

Data Science Intern

Jul 2025 – Aug 2025

- Engineered a scalable data collection pipeline using Python, leveraging Playwright and search APIs to scrape and process over 1000 resumes within a week, while ensuring regulatory compliance through robust data modeling and warehousing techniques.
- Developed a machine learning-based document classification system utilizing Sentence-BERT embeddings and Random Forest models to achieve approximately 90% accuracy in data classification, while applying data preprocessing techniques for enhanced data quality and integrity within an enterprise data analytics framework.
- Designed an end-to-end ETL workflow in Jupyter Notebook with SQLite caching to support data warehousing and enterprise data analytics, automated feature extraction for statistical analysis, and structured business intelligence dashboards for reporting key performance indicators (KPIs) on candidate sourcing insights.

The Data Mine Corporate Partners

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2022 – Apr 2023

- Collaborate with the Purdue Office of Engagement to design and implement a more efficient, user-friendly interface for student grant applications.
- Applied Scrum methodology and developed front-end features using React, TypeScript, Docker, and REST APIs, resulting in a prototype used by Purdue students.

CLASS AND PERSONAL PROJECTS

AI-Driven Dynamic Pricing for Autonomous Delivery

2026

- Built predictive models using structured data to simulate real-world decision systems, applying machine learning techniques for demand estimation and business intelligence-driven pricing strategy optimization in Microsoft Excel and Power BI environments.
- Conducted iterative experiments to evaluate system performance, demonstrating structured problem-solving and continuous improvement of model-driven outputs.
- Implemented and compared static and adaptive pricing strategies using Data Analysis techniques to evaluate revenue growth, customer wait time optimization, and congestion tradeoffs in a simulated environment, resulting in informed business decisions for Enterprise Data Analytics.

AI Story-to Video Automation Pipeline

2025

- Developed a fully automated AI-powered application that transforms raw text into video by integrating Large Language Model-driven data processing and statistical analysis for descriptive and inferential statistics, utilizing Microsoft Excel for data modeling and Power BI for business intelligence and data visualization, resulting in enhanced key performance indicators (KPIs) and key risk indicators (KRIs).
- Designed modular pipeline components and REST-based workflows to manage sequential transformations, utilizing Data Modeling and Business Intelligence principles for scalable system design, API integration, and orchestration in Power BI and Smartsheets environments.

Object Detection, OCR, and Grade Prediction for ER Diagrams

2024

- Developed and implemented a machine learning-based object detection module in Entity Relation Diagrams (ERDs) to accurately identify and classify components, utilizing Data Modeling techniques and Statistical Analysis methods for precise classification results.
- Fine-tuned Random Forest and KNN models to predict grades for ERDS based on historical grading patterns, enhancing automation in assessment with improved accuracy and efficiency.

LEADERSHIP AND INVOLVEMENT

Purdue Filipino Association

Aug 2022 – Present

President

- Developed and executed comprehensive event planning strategies for 200+ students, successfully leading a team of 20+ members to deliver cultural, social, and philanthropic events that fostered engagement and community building through Business Intelligence-driven insights and Data Visualization tools.
- Presented data-driven plans to board members during large-scale events, utilizing Business Intelligence and Data Visualization skills to inform strategic decisions, while demonstrating strong communication, organization, and client-facing abilities.

Asian Student Union Board

Aug 2023 – May 2025

Vice President Internal

- Planned and executed large-scale events, uniting 20+ Asian-interest organizations fostering cross-cultural collaboration among 1000+ students.
- Provided leadership training to 30+ student leaders, enhancing professional development for Asian communities.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C#, JavaScript, TypeScript, Bash

ML: Pandas, NumPy, Scikit-learn, Random Forest, KNN, EasyOCR

Data: Data Analysis, Data Validation, Data Cleaning, Trend Analysis, Statistical Analysis, Data Visualization
Tools & Infrastructure: Git, Docker, Playwright, FFmpeg, VS Code, Terminal, Jupyter Notebook, Databricks